

Linear differential systems

Let $n \in \mathbb{N}^*$, $A \in C(I, \mathcal{M}_n(\mathbb{R}))$ where $I \subset \mathbb{R}$ open interval, $F \in C(I, \mathbb{R}^n)$.
we consider the system

$$(1) \quad X' = A(t)X + F(t)$$

this is a linear non-hom (when $F \neq 0$) differential system in $\mathbb{R}^n$.

Here $A(t)$ is the coefficients matrix, $F(t)$ is the non-hom. part of (1),
while $X' = A(t)X$ is the homogeneous part of (1).

Def 1 A solution of (1) is a function $\varphi: I \rightarrow \mathbb{R}^n$ of class C^1 such that
 $\varphi'(t) = A(t)\varphi(t) + F(t)$, $\forall t \in I$.

Now we write the system by components

$$A(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}(t) & a_{m2}(t) & \dots & a_{nn}(t) \end{pmatrix}$$

$$F(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix} \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$(1) \Leftrightarrow \begin{cases} x_1' = a_{11}(t)x_1 + a_{12}(t)x_2 + \dots + a_{1n}(t)x_n + f_1(t) \\ x_2' = a_{21}(t)x_1 + a_{22}(t)x_2 + \dots + a_{2n}(t)x_n + f_2(t) \\ \vdots \\ x_n' = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \dots + a_{nn}(t)x_n + f_n(t) \end{cases}$$

Def 2. A matrix solution of $X' = A(t)X$ is a function $U: I \rightarrow \mathcal{M}_n(\mathbb{R})$ of class C^1 s.t.
 $U'(t) = A(t)U(t)$, $\forall t \in I$.

Property A matrix function $U: I \rightarrow \mathcal{M}_n(\mathbb{R})$, of class C^1 , is a matrix solution of $X' = A(t)X$ if and only if each column of $U(t)$ is a solution of $X' = A(t)X$.

Theorem 1. (The existence and uniqueness theorem for the IVP).

Let $t_0 \in I$ and $\eta \in \mathbb{R}^n$. The IVP $\begin{cases} X' = A(t)X + F(t) \\ X(t_0) = \eta \end{cases}$ has a unique solution.

Definition The matrix function $E: \mathbb{R} \rightarrow \mathcal{M}_n(\mathbb{R})$ which is the unique

Definition The matrix function $E: \mathbb{R} \rightarrow \mathcal{M}_m(\mathbb{R})$ which is the unique matrix function such that $\begin{cases} E'(t) = A E(t) \\ E(0) = I_m \end{cases}$ is said to be the exponential of the matrix A and is denoted by $E(t) = e^{tA}$ or $E(t) = \exp(tA)$.

Here $A \in \mathcal{M}_m(\mathbb{R})$ and $I_m = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ is the identity matrix.

Property Let $A \in \mathcal{M}_m(\mathbb{R})$. The series

$$I_m + A + \frac{1}{2!} A^2 + \dots + \frac{1}{k!} A^k + \dots$$

is convergent to e^A .

Thus, $\left[e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \right]$ (an equivalent definition of the matrix exponential)

Examples. Compute $e^{t \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}}$ in two ways

where $\lambda_1, \lambda_2 \in \mathbb{R}$.

Sol $E(t) = e^{tA}$, where

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

First we use the definition of e^{tA} .

Step 1 we write the LHS with matrix A

$$X' = AX \Leftrightarrow \begin{cases} x_1' = \lambda_1 x_1 \\ x_2' = \lambda_2 x_2 \end{cases}$$

this is an uncoupled system, so it is easy to solve it.

$$x_1' = \lambda_1 x_1 \Leftrightarrow \begin{cases} x_1' - \lambda_1 x_1 = 0 \\ \lambda_2 - \lambda_1 = 0 \\ \lambda_2 = \lambda_1 \mapsto e^{\lambda_1 t} \end{cases} \Rightarrow \begin{cases} x_1 = c_1 e^{\lambda_1 t} \\ c_1 \in \mathbb{R} \\ x_2 = c_2 e^{\lambda_2 t} \\ c_2 \in \mathbb{R} \end{cases}$$

Step 2 we formulate 2 ivp's

$$(2) \begin{cases} x_1' = \lambda_1 x_1 \\ x_2' = \lambda_2 x_2 \\ x_1(0) = 1 \\ x_2(0) = 0 \end{cases} \text{ has the unique solution } \varphi(t) = \begin{pmatrix} e^{\lambda_1 t} \\ 0 \end{pmatrix}$$

$$(3) \begin{cases} x_1' = \lambda_1 x_1 \\ x_2' = \lambda_2 x_2 \\ x_1(0) = 0 \\ x_2(0) = 1 \end{cases} \text{ has the unique solution } \psi(t) = \begin{pmatrix} 0 \\ e^{\lambda_2 t} \end{pmatrix}$$

$$\text{Step 3. } E(t) = \begin{pmatrix} \text{sol. of (2)} & \text{sol. of (3)} \end{pmatrix} = \begin{pmatrix} \varphi(t) & \psi(t) \end{pmatrix} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}$$

$$\text{Conclusion } e^{t \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}, \quad \forall t \in \mathbb{R}.$$

Conclusion $e^{t \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}, \forall t \in \mathbb{R}.$

Now we use the equivalent definition with Taylor series.

$$e^{tA} = I_2 + tA + \frac{1}{2!}(tA)^2 + \dots + \frac{1}{k!}(tA)^k + \dots = I_2 + tA + \frac{t^2}{2!}A^2 + \dots + \frac{t^k}{k!}A^k + \dots$$

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad A^2 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{pmatrix} \dots$$

By induction one can prove that $A^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix} \quad \forall k \in \mathbb{N}$

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k}{k!} \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix} = \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_1 t)^k & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_2 t)^k \end{pmatrix} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}$$

Linear homogeneous systems of dimension 2 with constant coefficients
solved using the reduction method.

$$(4) \quad \begin{cases} x' = a_{11}x + a_{12}y \\ y' = a_{21}x + a_{22}y \end{cases} \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in \mathcal{M}_n(\mathbb{R})$$

we assume that (4) is a coupled system, i.e. either $a_{12} \neq 0$ or $a_{21} \neq 0$.

To fix, consider that $a_{12} \neq 0$. Our aim is to find a second order LDE for x , i.e.

a relation between x, x' and x'' . (Comment: if $a_{12} = 0$ but $a_{21} \neq 0$ we find a relation between y, y', y'')

$$x' = a_{11}x + a_{12}y \Rightarrow y = \frac{1}{a_{12}}(x' - a_{11}x) \quad (5)$$

$$x' = a_{11}x + a_{12}y \Rightarrow x'' = a_{11}x' + a_{12}y' \Rightarrow x'' = a_{11}x' + a_{12} \cdot (a_{21}x + a_{22}y) \Rightarrow \quad (5)$$

$\uparrow$
 $y' = a_{21}x + a_{22}y$

$$\Rightarrow x'' = a_{11}x' + a_{12}a_{21}x + a_{22}(x' - a_{11}x) \Rightarrow$$

$$\Rightarrow x'' - (a_{11} + a_{22})x' + (a_{11}a_{22} - a_{12}a_{21})x = 0 \quad (6) \text{ this is a second order LDE for } x$$

$\Rightarrow x'' - (a_{11} + a_{22})x' + (a_{11}a_{22} - a_{12}a_{21})x = 0$ (6) this is a second order LHDE with constant coefficients that can be solved using the characteristic equation method.

Remarks. The characteristic polynomial of (6) is $l(r) = r^2 - \underbrace{(a_{11} + a_{22})}_{\text{tr } A} r + \underbrace{(a_{11}a_{22} - a_{12}a_{21})}_{\det A}$

$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. Note that $l(r)$ is the characteristic polynomial of A .

Recall that the roots of $l(r)$ are the eigenvalues of A .

Recall the definition of an eigenvalue and an eigenvector of a matrix $A \in M_n(\mathbb{R})$:

we say that $\lambda \in \mathbb{C}$ is an eigenvalue of the matrix A when

$\exists u \in \mathbb{R}^n$, $u \neq 0$ such that $Au = \lambda u$. This u is said to be an eigenvector of A corresp. to the eigenvalue λ .

Example. Compute in two ways $e^{t \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}$, $t \in \mathbb{R}$.

Method I. $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ the LHDS having A as the coeff. matrix is

Step 1
 $\begin{cases} x' = y \\ y' = -x \end{cases}$ this is a coupled system. we use the reduction method to find its general sol.

$$x'' = y' \Rightarrow x'' = -x \Rightarrow x'' + x = 0$$

$$y = x'$$

$$r^2 + 1 = 0 \quad r^2 = -1 \quad r_{1,2} = \pm i \mapsto \cos t, \sin t$$

$$(\alpha \pm i\beta \mapsto e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t)$$

$$\begin{cases} x = c_1 \cos t + c_2 \sin t \\ y = -c_1 \sin t + c_2 \cos t \end{cases}, c_1, c_2 \in \mathbb{R}$$

Step 2

$$\begin{cases} x' = y \\ y' = -x \\ x(0) = 1 \\ y(0) = 0 \end{cases}$$

$$\begin{aligned} x &= c_1 \cos t + c_2 \sin t \\ y &= -c_1 \sin t + c_2 \cos t \end{aligned}$$

$$x(0) = c_1 \Rightarrow c_1 = 1 \text{ and } c_2 = 0$$

$$y(0) = c_2$$

the unique sol. of this IVP is $\varphi(t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$

$$\begin{cases} x' = y \\ y' = -x \end{cases}$$

$$c_1 = 0$$

the unique sol. of this IVP is $\psi(t) = \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$

$$\begin{cases} x' = y \\ y' = -x \\ x(0) = 0 \\ y(0) = 1 \end{cases} \quad \begin{matrix} C_1 = 0 \\ C_2 = 1 \end{matrix} \quad \text{the unique sol. of this IVP is } \psi(t) = \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

Step 3

$$e^{t \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}, \quad \forall t \in \mathbb{R}$$

Method II HW